

Lecture 5: RBC with Endogenous Labor

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HP filter

HP filter: decompose economic series into trend and cyclical component

$$y_t = y_t^c + y_t^T$$

$$\min_{\{y_t^T\}_{t=1}^N} \mathcal{L} = \underbrace{\sum_{t=1}^N (y_t - y_t^T)^2}_{(1)} + \lambda \underbrace{\sum_{t=2}^{N-1} \left[(y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T) \right]^2}_{(2)}$$

Notes:

- 1) penalty of calling everything trend
- 2) change in trend growth rate, penalty of wiggles

If λ huge, close to straight line in logs.

If $\lambda = 0$, wiggle a lot, fit to y_t .

Define

$$\Delta^2 y_t^T \equiv y_{t+1}^T - 2y_t^T + y_{t-1}^T \quad (\text{second-order difference})$$

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = 0 \quad \Rightarrow \quad \frac{\partial (1)}{\partial y_t^T} = 2(y_t^T - y_t)$$

Since

$$\Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T \quad (\text{coefficient } +1)$$

$$\Delta^2 y_t^T = y_{t+1}^T - 2y_t^T + y_{t-1}^T \quad (\text{coefficient } -2)$$

$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T \quad (\text{coefficient } +1)$$

$$\begin{aligned} \frac{\partial}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 &= 2\Delta^2 y_{t-1}^T \frac{\partial \Delta^2 y_{t-1}^T}{\partial y_t^T} + 2\Delta^2 y_t^T \frac{\partial \Delta^2 y_t^T}{\partial y_t^T} + 2\Delta^2 y_{t+1}^T \frac{\partial \Delta^2 y_{t+1}^T}{\partial y_t^T} \\ &= 2(\Delta^2 y_{t-1}^T - 2\Delta^2 y_t^T + \Delta^2 y_{t+1}^T) \end{aligned}$$

Since

$$\begin{aligned} \Delta^2 y_{t-1}^T &= y_t^T - 2y_{t-1}^T + y_{t-2}^T, \\ -2\Delta^2 y_t^T &= -2(y_{t+1}^T - 2y_t^T + y_{t-1}^T) = -2y_{t+1}^T + 4y_t^T - 2y_{t-1}^T, \end{aligned}$$

$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T,$$

we get

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = \frac{\partial(1)}{\partial y_t^T} + \lambda \frac{\partial}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 = 0.$$

$$\Rightarrow (1 + 6\lambda)y_t^T - 4\lambda y_{t-1}^T - 4\lambda y_{t+1}^T + \lambda y_{t-2}^T + \lambda y_{t+2}^T = y_t^T, \quad 3 \leq t \leq N-2.$$

At $t = 1$ (only $\Delta^2 y_2^T$ exists),

$$(1 + \lambda)y_1^T - 2\lambda y_2^T + \lambda y_3^T = y_1^T.$$

At $t = 2$ (only $\Delta^2 y_2^T$ and $\Delta^2 y_3^T$ exist),

$$-2\lambda y_1^T + (1 + 5\lambda)y_2^T - 4\lambda y_3^T + \lambda y_4^T = y_2^T.$$

Symmetrically for $t = N$ and $t = N - 1$, collect $\{y_t\}$ into the $N \times 1$ vector y and $\{y_t^T\}$ into y^T , then we have

$$Ay^T = y \quad \Rightarrow \quad y^T = A^{-1}y \quad \Rightarrow \quad y^c = y - A^{-1}y.$$

where

$$A = \begin{pmatrix} 1 + \lambda & -2\lambda & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 \\ -2\lambda & 1 + 5\lambda & -4\lambda & \lambda & 0 & \cdots & 0 & 0 & 0 \\ \lambda & -4\lambda & 1 + 6\lambda & -4\lambda & \lambda & \cdots & 0 & 0 & 0 \\ 0 & \lambda & -4\lambda & 1 + 6\lambda & -4\lambda & \ddots & 0 & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -4\lambda & 1 + 6\lambda & -4\lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda & -4\lambda & 1 + 5\lambda & -2\lambda \\ 0 & 0 & 0 & \cdots & 0 & 0 & \lambda & -2\lambda & 1 + \lambda \end{pmatrix}.$$

Let D be the $(N - 2) \times N$ second-difference matrix, whose t -th row ($t = 2, \dots, N - 1$) looks like

$$[0 \cdots 1 \ -2 \ 1 \ \cdots 0]$$

so that

$$(Dy^T)_{t-1} = \Delta^2 y_t^T.$$

The HP objective becomes

$$\min_{y^T} (y - y^T)'(y - y^T) + \lambda(Dy^T)'(Dy^T).$$

F.O.C.

$$-2(y - y^T) + 2\lambda D'Dy^T = 0 \quad \Rightarrow \quad (I + \lambda D'D)y^T = y.$$

Hence

$$A = I + \lambda D'D.$$

As a 5-diagonal pattern:

$$\text{interior diagonal: } [0 \cdots 1 \ -2 \ 1 \ \cdots 0] \begin{bmatrix} 0 \\ \vdots \\ 1 \\ -2 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 6$$

$$\text{first-off diagonal: } [0 \cdots 1 \ -2 \ 1 \ 0 \cdots 0] \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = -4$$

$$\text{second-off diagonal: } [0 \cdots 1 \ -2 \ 1 \ 0 \ 0 \cdots 0] \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1$$

$\lambda \uparrow \Rightarrow$ the smoother the series y_t^T .

Pros

- 1) Simple and transparent. (defined by one objective)
- 2) Produces a smooth trend with tunable smoothness λ
- 3) Widely used in Macro

Cons

- 1) End-point problem: start and end of sample is distorted.
- 2) λ sensitivity: moments (volatility, correlations) change a lot with λ .
- 3) Spurious dynamics: HP can induce artificial serial correlations and distort cross-corr between series (create cycles).
- 4) Leakage: low-frequency cycles leak into trend.
- 5) Not structural: no model interpretation.

Why $\lambda = 1600$ is the quarterly standard?

Rule of thumb:

$$\lambda = 100 \times (\text{periods per year})^2.$$

Hence

$$\lambda_{\text{quarter}} = 100 \times 4^2 = 1600, \quad \lambda_{\text{month}} = 100 \times 12^2 = 14400, \quad \lambda_{\text{annual}} = 100 \times 1^2 = 100.$$

Long and Plosser (1983, JPE)

Social planner's Problem Setup:

$$U = \max_{\{C_t, L_t, K_t\}} E_0 \left\{ \sum_{t=0}^{\infty} \beta^t [\theta \ln C_t + (1 - \theta) \ln L_t] \right\}$$

s.t.

$$K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} - C_t \quad \text{and } K_0 \text{ is given.}$$

Resource constraint:

$$C_t + K_t = Y_t.$$

$$N_t + L_t = 1 \quad (\text{trade-off})$$

L : leisure N : working hour.

Total time is normalized to 1.

where

$$\log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2), \quad \psi < 1.$$

Note: capital completely turns over,

$$(1 - \delta)K_{t-1} = 0 \quad \delta = 1.$$

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\theta \ln C_t + (1 - \theta) \ln(1 - N_t) + \lambda_t (Z_t K_{t-1}^\alpha N_t^{1-\alpha} - C_t - K_t) \right].$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \frac{\theta}{C_t} = \lambda_t \quad (1)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad &\Rightarrow \quad -\underbrace{\frac{1 - \theta}{1 - N_t}}_{\text{utility loss}} + \underbrace{\lambda_t}_{\text{MUC}_t} \underbrace{(1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}}_{\text{MPN}_t} = 0 \\ &\Rightarrow \quad \frac{1 - \theta}{1 - N_t} = (1 - \alpha) \lambda_t Z_t K_{t-1}^\alpha N_t^{-\alpha} \end{aligned} \quad (2)$$

labor-leisure condition (intratemporal)

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad &\Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t \left[\lambda_{t+1} \frac{\partial}{\partial K_t} (Z_{t+1} K_t^\alpha N_{t+1}^{1-\alpha}) \right] = 0 \\ &\Rightarrow \quad \lambda_t = \beta E_t (\lambda_{t+1} \alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha}) \end{aligned} \quad (3)$$

Euler equation (intertemporal)

TVC:

$$\lim_{T \rightarrow \infty} \beta^T U_C(C, L) K_{T+1} = 0. \quad (4)$$

Closed-form solution: constant savings rate and constant labor

Tractability: log utility + Cobb–Douglas + full depreciation

 $\Rightarrow$ constant savings rate / hours.

Since

$$Y_{t+1} = Z_{t+1} K_t^\alpha N_{t+1}^{1-\alpha},$$

we have

$$\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} = \alpha \frac{Y_{t+1}}{K_t}.$$

Plug into (3):

$$\lambda_t = \beta E_t \left(\lambda_{t+1} \alpha \frac{Y_{t+1}}{K_t} \right).$$

Using (1):

$$\begin{aligned} \frac{\theta}{C_t} &= \beta E_t \left(\frac{\theta}{C_{t+1}} \alpha \frac{Y_{t+1}}{K_t} \right) \\ \Rightarrow \frac{1}{C_t} &= \beta \alpha E_t \left[\frac{Y_{t+1}}{C_{t+1}} \frac{1}{K_t} \right]. \end{aligned} \quad (5)$$

Guess and verify:

$$C_t = (1-s)Y_t, \quad s \text{ is the savings rate}, \quad K_t = sY_t. \quad (\text{full depreciate})$$

Then

$$\frac{Y_{t+1}}{C_{t+1}} = \frac{1}{1-s}.$$

Plug into (5):

$$\begin{aligned} \frac{1}{(1-s)Y_t} &= \alpha \beta \frac{1}{1-s} \frac{1}{sY_t} \\ \Rightarrow s &= \alpha \beta. \end{aligned}$$

Hence

$$C_t = (1 - \alpha\beta)Y_t, \quad K_t = \alpha\beta Y_t.$$

$$\begin{cases} \alpha \uparrow \Rightarrow \text{capital income share} \uparrow \Rightarrow \text{savings rate} \uparrow, \\ \beta \uparrow \Rightarrow \text{discount factor} \downarrow \Rightarrow \text{savings rate} \uparrow. \end{cases}$$

Plug into (2):

$$\begin{aligned} \frac{1-\theta}{1-N_t} &= (1-\alpha) \frac{\theta}{C_t} \frac{Y_t}{N_t} = (1-\alpha) \frac{\theta}{(1-\alpha\beta)Y_t} \frac{Y_t}{N_t} \\ \Rightarrow (1-\theta)(1-\alpha\beta)N_t &= (1-\alpha)\theta(1-N_t) \\ \Rightarrow N_t = \bar{N} &= \frac{\theta(1-\alpha)}{\theta(1-\alpha) + (1-\theta)(1-\alpha\beta)}. \end{aligned}$$

Since $N_t = \text{constant } \forall t$, we have

$$\hat{n}_t = 0.$$

Also

$$K_t = \alpha\beta Y_t, \quad C_t = (1 - \alpha\beta)Y_t$$

so

$$\hat{k}_t = \hat{c}_t = \hat{y}_t.$$

Since

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1} + (1 - \alpha)\hat{n}_t = \hat{z}_t + \alpha\hat{k}_{t-1},$$

we get

$$\hat{k}_t = \hat{c}_t = \hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1}.$$

And

$$\hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t.$$

TFP shock alone can generate output/consumption/investment fluctuations, but

- 1) labor does not fluctuate at all,
- 2) c, k, y move one-for-one.

Consumption is usually smoother than output; capital is usually more volatile.

- 3) Labor productivity moves like output in model.

$$\left(\frac{\hat{Y}}{\hat{N}} \right)_t = \hat{y}_t - \hat{n}_t = \hat{y}_t, \quad \text{while in data,}$$

$$\sigma(Y/N) \ll \sigma(Y), \quad \text{and } \text{Corr}(Y/N, Y) = 0.55.$$

- 4) Prices (MRS, MRT, MPL, MPK) too-tight with output & TFP, hard to generate weak wage cyclical & interest rate counter-cyclical.

Labor market equilibrium

Household's problem:

$$U_{N,t} + \lambda_t W_t = 0 \quad \Rightarrow \quad U_{N,t} + U_{C,t} W_t = 0$$

labor supply

Firm's problem:

$$\max_{K_{t-1}, N_t} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

labor demand

Hence

$$W_t = Z_t \frac{Y_t}{N_t} (1 - \alpha).$$

From household labor supply:

$$W_t = \frac{1 - \theta}{\theta} \frac{C_t}{1 - N_t}.$$

Equivalently,

$$W_t = \frac{1 - \theta}{\theta} \frac{(1 - \alpha\beta)Z_t K_{t-1}^\alpha N_t^{1-\alpha}}{1 - N_t}.$$

From firm labor demand:

$$W_t = (1 - \alpha)Z_t K_{t-1}^\alpha N_t^{-\alpha}.$$

A positive Z_t or K_{t-1} shock shifts labor supply up by the same factor as labor demand:

$$Z_t K_{t-1}^\alpha.$$

Hence labor supply's wealth effect is exactly offset by labor demand increase. So when Z_t rises, the real wage increases. This generates a substitution effect toward more labor, but also an income effect toward more leisure.

$$W_t = (1 - \alpha)Z_t K_{t-1}^\alpha N_t^{-\alpha}$$

$$Z_t \uparrow \Rightarrow MPN_t \uparrow \Rightarrow W_t \uparrow \Rightarrow \text{labor demand curve shifts up}$$

$$Z_t \uparrow \Rightarrow Y_t \uparrow \Rightarrow C_t = (1 - \alpha\beta)Y_t \uparrow \Rightarrow MU_{C,t} \downarrow \Rightarrow N_t \downarrow \quad \text{Wealth effect}$$

$$\begin{aligned} \underbrace{\frac{1 - \theta}{1 - N_t}}_{\text{marginal disutility of working}} &= \underbrace{\lambda_t W_t}_{\text{marginal utility value of wage income}} = \underbrace{MU_{C,t} W_t}_{\text{intra-temporal optimality}} = \frac{\theta}{C_t} (1 - \alpha) \frac{Y_t}{N_t} \\ &= \frac{\theta(1 - \alpha)}{N_t} \left(\frac{Y_t}{C_t} \right) = \frac{\theta(1 - \alpha)}{1 - \alpha\beta} \frac{1}{N_t}. \end{aligned}$$

Hansen (1985): Basic idea Expand log preference to make labor respond to shocks.
Household's problem:

$$\max_{\{C_t, L_t, N_t\}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t) = E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - N_t)$$

$$\text{s.t.} \quad N_t + L_t = 1$$

$$C_t + K_t = W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1} \quad (\text{budget constraint not resource constraint})$$

$$K_t - (1 - \delta) K_{t-1} = I_t \quad \text{investment}$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[u(C_t, 1 - N_t) + \lambda_t (W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1} - C_t - K_t) \right]$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \underbrace{U_{C,t}}_{\text{marginal utility of consumption}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad U_{N,t} + \lambda_t W_t = 0$$

$$\Rightarrow \quad - \underbrace{U_{N,t}}_{\text{marginal disutility of working}} = \underbrace{\lambda_t W_t}_{\text{marginal utility value of one more unit of wage income}}$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad \Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t [\lambda_{t+1} R_{t+1} + \lambda_{t+1} (1 - \delta)] = 0$$

$$\Rightarrow \quad U_{C,t} = \beta E_t [U_{C,t+1} (R_{t+1} + 1 - \delta)]$$

Firm's problem

$$\max_{N_t, K_{t-1}} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

s.t.

$$Y_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha}, \quad \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t.$$

FOCs:

$$\frac{\partial \Pi_t}{\partial K_{t-1}} = 0 \quad \Rightarrow \quad \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} = R_t \quad \Rightarrow \quad \alpha \frac{Y_t}{K_{t-1}} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t} = 0 \quad \Rightarrow \quad (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha} = W_t \quad \Rightarrow \quad (1 - \alpha) \frac{Y_t}{N_t} = W_t.$$

Let utility function follow

$$U(C_t, 1 - N_t) = \ln C_t + A \ln(1 - N_t).$$

Then

$$U_{C,t} = \frac{1}{C_t}, \quad U_{N,t} = -\frac{A}{1 - N_t}.$$

Hence equilibrium conditions are

$$\left\{ \begin{array}{l} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (R_{t+1} + 1 - \delta) \right], \\ \frac{A}{1 - N_t} = \frac{1}{C_t} W_t, \\ W_t = (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}, \\ R_t = \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha}, \\ \Pi_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} - W_t N_t - R_t K_{t-1} = 0, \\ C_t + K_t = W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1}, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{array} \right.$$

Do not substitute out R_t, W_t , to make a system of equations of (C_t, N_t, K_t, Z_t) .
Substituting yields

$$\left\{ \begin{array}{l} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right], \\ A C_t = (1 - N_t) (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}, \\ K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{array} \right.$$

Step 1: Steady state $\Rightarrow$ Step 2: Log-linearize $\Rightarrow$ Step 3: Numerically solve

Steady state

At steady state:

$$K_t = K_{t-1} = \bar{K}, \quad C_t = C_{t+1} = \bar{C}, \quad N_t = N_{t+1} = \bar{N}.$$

$$\bar{K} = \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} + (1 - \delta) \bar{K} - \bar{C} \Rightarrow \bar{C} = \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K}. \quad (1)$$

$$1 = \alpha \beta \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} + (1 - \delta) \beta. \quad (2)$$

$$A \bar{C} = (1 - \bar{N})(1 - \alpha) \bar{Z} \bar{K}^\alpha \bar{N}^{-\alpha}. \quad (3)$$

$$\Rightarrow \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} = \frac{1}{A} \left((1 - \alpha) \bar{Z} \bar{K}^\alpha \bar{N}^{-\alpha} - (1 - \alpha) \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} \right)$$

$$\Rightarrow \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} + \frac{1 - \alpha}{A} \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{1 - \alpha}{A} \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta$$

$$\Rightarrow \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{A}{1 - \alpha + A} \left[\frac{1 - \alpha}{A} \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta \right]$$

Using (2), after simplification one gets

$$\bar{N} = \left\{ 1 + \frac{A}{1 - \alpha} \left[1 - \frac{\alpha \beta \delta}{1 - \beta(1 - \delta)} \right] \right\}^{-1}.$$

Then

$$\bar{K} = \left[\frac{1 - \beta(1 - \delta)}{\bar{Z} \alpha \beta} \bar{N}^{\alpha-1} \right]^{\frac{1}{\alpha-1}} = \bar{N} \left[\frac{\bar{Z} \alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{1}{1-\alpha}}.$$

And

$$\begin{aligned} \bar{C} &= \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} = \bar{N}^\alpha \cdot \left[\frac{\bar{Z} \alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{\alpha}{1-\alpha}} \bar{N}^{1-\alpha} \bar{Z} - \delta \bar{K} \\ &= \bar{Z}^{\frac{1}{1-\alpha}} \bar{N} \left[\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{\alpha}{1-\alpha}} - \delta \bar{N} \bar{Z}^{\frac{1}{1-\alpha}} \left[\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{1}{1-\alpha}} \\ &= \bar{N} \bar{Z}^{\frac{1}{1-\alpha}} \left[\left(\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right)^{\frac{\alpha}{1-\alpha}} - \delta \left(\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right)^{\frac{1}{1-\alpha}} \right] \end{aligned}$$

Derive the log-linearized system

(i) TFP process:

$$\log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t$$

Let

$$Z_t = \bar{Z} e^{\hat{z}_t}.$$

Then

$$\begin{aligned} \hat{z}_t + \log \bar{Z} &= (1 - \psi) \log \bar{Z} + \psi \log \bar{Z} + \psi \hat{z}_{t-1} + \varepsilon_t \\ \Rightarrow \hat{z}_t &= \psi \hat{z}_{t-1} + \varepsilon_t. \end{aligned}$$

(ii)

$$K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta)K_{t-1} - C_t.$$

Substitute

$$K_t = \bar{K}(1 + \hat{k}_t), \quad Z_t = \bar{Z}(1 + \hat{z}_t), \quad K_{t-1} = \bar{K}(1 + \hat{k}_{t-1}), \quad N_t = \bar{N}(1 + \hat{n}_t), \quad C_t = \bar{C}(1 + \hat{c}_t),$$

and keep only first-order terms:

$$\bar{K} + \bar{K}\hat{k}_t = \bar{Z}\bar{K}^\alpha\bar{N}^{1-\alpha}(1 + \hat{z}_t)(1 + \alpha\hat{k}_{t-1})(1 + (1 - \alpha)\hat{n}_t) + (1 - \delta)\bar{K} + (1 - \delta)\bar{K}\hat{k}_{t-1} - \bar{C} - \bar{C}\hat{c}_t.$$

Using the steady state condition and dropping higher-order terms:

$$\bar{K}\hat{k}_t = \alpha\bar{Y}\hat{k}_{t-1} + \bar{Y}\hat{z}_t + (1 - \alpha)\bar{Y}\hat{n}_t + (1 - \delta)\bar{K}\hat{k}_{t-1} - \bar{C}\hat{c}_t,$$

where

$$\bar{Y} \equiv \bar{Z}\bar{K}^\alpha\bar{N}^{1-\alpha}.$$

Hence

$$\hat{k}_t = \left(\alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta \right) \hat{k}_{t-1} + (1 - \alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + \frac{\bar{Y}}{\bar{K}} \hat{z}_t - \frac{\bar{C}}{\bar{K}} \hat{c}_t.$$

(iii)

$$AC_t = (1 - N_t)(1 - \alpha)Z_t K_{t-1}^\alpha N_t^{-\alpha}.$$

Log-linearizing gives

$$\hat{n}_t = \left(\alpha + \frac{\bar{N}}{1 - \bar{N}} \right)^{-1} (\hat{z}_t + \alpha\hat{k}_{t-1} - \hat{c}_t).$$

(iv) Euler equation:

$$1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right].$$

Let

$$M_{t+1} \equiv \alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha}, \quad G_{t+1} \equiv M_{t+1} + 1 - \delta.$$

Then

$$\hat{m}_{t+1} = \hat{z}_{t+1} + (\alpha - 1)\hat{k}_t + (1 - \alpha)\hat{n}_{t+1}.$$

Define

$$\tilde{\delta} \equiv \frac{\bar{M}}{\bar{G}}.$$

Since

$$\bar{G} = \beta^{-1}, \quad \bar{G} = \bar{M} + (1 - \delta),$$

we have

$$\tilde{\delta} = \frac{\bar{M}}{\bar{G}} = 1 - \frac{1 - \delta}{\bar{G}} = 1 - \beta(1 - \delta).$$

Thus

$$G_{t+1} \approx \bar{G}(1 + \tilde{\delta}\hat{m}_{t+1}).$$

Also

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C}e^{\hat{c}_t}}{\bar{C}e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}} \approx 1 + \hat{c}_t - \hat{c}_{t+1}.$$

So Euler equation becomes

$$1 = \beta E_t \left[\bar{G}(1 + \hat{c}_t - \hat{c}_{t+1} + \tilde{\delta} \hat{m}_{t+1}) \right]$$

Since

$$\beta \bar{G} = 1,$$

we obtain

$$E_t [\hat{c}_t - \hat{c}_{t+1} + \tilde{\delta} \hat{m}_{t+1}] = 0.$$

Hence

$$\hat{c}_t = E_t \hat{c}_{t+1} - \tilde{\delta} E_t [\hat{z}_{t+1} - (1 - \alpha) \hat{k}_t + (1 - \alpha) \hat{n}_{t+1}].$$

If you have CRRA,

$$\frac{C_t}{C_{t+1}} \rightarrow \left(\frac{C_t}{C_{t+1}} \right)^\sigma \Rightarrow \sigma \hat{c}_t = \sigma E_t \hat{c}_{t+1} - E_t \hat{r}_{t+1}.$$

Thus we have a system of equations:

$$\begin{cases} \hat{z}_t = \psi \hat{z}_{t-1} + \varepsilon_t, \\ \hat{c}_t = E_t \hat{c}_{t+1} - \tilde{\delta} E_t [\hat{z}_{t+1} - (1 - \alpha) \hat{k}_t + (1 - \alpha) \hat{n}_{t+1}], \\ \hat{n}_t = \left(\alpha + \frac{\bar{N}}{1 - \bar{N}} \right)^{-1} (\hat{z}_t + \alpha \hat{k}_{t-1} - \hat{c}_t), \\ \hat{k}_t = \left(\alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta \right) \hat{k}_{t-1} + (1 - \alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + \frac{\bar{Y}}{\bar{K}} \hat{z}_t - \frac{\bar{C}}{\bar{K}} \hat{c}_t. \end{cases}$$

Four equations, four variables:

$$(\hat{k}_{t-1}, \hat{c}_t, \hat{n}_t, \hat{z}_t).$$

$\hat{k}_{t-1}$ is predetermined variable, $\hat{z}_t$ follows a stable AR(1) process,

$(\hat{c}_t, \hat{n}_t)$ are jump variables.

Use undetermined coefficient method; guess

$$\hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \hat{z}_t, \quad \hat{c}_t = V_{ck} \hat{k}_{t-1} + V_{cz} \hat{z}_t, \quad \hat{n}_t = V_{nk} \hat{k}_{t-1} + V_{nz} \hat{z}_t.$$

Then

$$\hat{k}_t = V_{kz} \sum_{i=0}^{\infty} \sum_{j=0}^i V_{kk}^j \psi^{i-j} \varepsilon_{t-i}, \quad \hat{z}_t = \sum_{s=0}^{\infty} \psi^s \varepsilon_{t-s},$$

and since

$$\hat{y}_t = \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{n}_t + \hat{z}_t,$$

we can write

$$\hat{y}_t = V_{yz} \varepsilon_t + \sum_{i=0}^{\infty} \left(V_{yk} V_{kz} \sum_{j=0}^i V_{kk}^j \psi^{i-j} + V_{yz} \psi^{i+1} \right) \varepsilon_{t-i-1}.$$

Therefore,

$$\text{Var}(\hat{y}_t) = \left(V_{yz}^2 + \sum_{i=0}^{\infty} \left[V_{yk} V_{kz} \sum_{j=0}^i V_{kk}^j \psi^{i-j} + V_{yz} \psi^{i+1} \right]^2 \right) \text{Var}(\varepsilon_t).$$

Given parameters' value, we can back out the explicit numerical solution.
This model works:

- 1) well for consumption volatility
- 2) bad for output volatility: requires too small TFP volatility
- 3) bad for hours volatility $\Rightarrow$ way too low.
- 4) bad for investment volatility $\Rightarrow$ far too low.

But it allows N_t to move with shocks compared to Long–Plosser. While with this set of preference and technology, it cannot generate enough labor and investment volatility relative to output.

Hansen (1985) with indivisible labor

Labor market phenomena:

- 1) Hours fluctuate a lot relative to productivity.
- 2) Micro evidence does not support huge intertemporal substitution of labor at individual level.

Target: get big aggregate hours fluctuations without assuming each person is very willing to shift leisure across time.

Key mechanism:

- 1) Each individual's labor is a corner-choice:

$$\tilde{n}_t \in \{0, n^*\}.$$

Either work a fixed shift or no work.

- 2) Lottery over jobs: assigned a job with probability π_t .
- 3) Complete unemployment insurance: C_t is independent of job status.

Crucial: we can write expected utility clearly at the “representative agent” level.

Thus total labor supply:

$$H_t = \bar{L} \pi_t n^*$$

where $\bar{L}$ is total labor force.

Average labor supply:

$$n_t = \frac{H_t}{\bar{L}} = \pi_t n^*.$$

Representative-agent utility: individual utility is

$$U(C_t) + v(\tilde{n}_t).$$

Given lottery,

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t v(n^*) + (1 - \pi_t)v(0).$$

Let

$$v(0) = 0, \quad \frac{v(n^*)}{n^*} = -A < 0.$$

Then

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t(-An^*) = U(C_t) - An_t.$$

Suppose

$$U(C_t) = \ln C_t \quad \Rightarrow \quad U_t = \ln C_t - An_t.$$

Hence

$$U_{C,t} = \frac{1}{C_t}, \quad U_{N,t} = -A, \quad -U_{N,t} = U_{C,t} W_t$$

Micro-level: no smoothing in labor, either n^* or 0, keep small intertemporal labor substitution.

Macro-level: planner / representative agent can vary π_t a lot, aggregate hours become very responsive.

Thus, we can define the equilibrium conditions for this model:

$$\begin{cases} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right], \\ K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t, \\ AC_t = (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}, \quad \text{key difference} \quad - U_{N,t} = U_{C,t} W_t = U_{C,t} MPL_t, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{cases}$$

Mechanism

In the indivisible labor model, aggregate hours fluctuate mainly through the employment margin: more people work rather than each individual continuously adjusting hours.

By contrast, in the divisible labor model,

$$TFP \uparrow \Rightarrow \text{demand for labor} \uparrow \Rightarrow W_t \uparrow \Rightarrow N_t \uparrow \Rightarrow 1 - N_t \downarrow$$

$\ln(1 - N_t)$ becomes steeper $\Rightarrow$ the cost of giving up a marginal hour of leisure rises

$\Rightarrow$ unwilling to adjust N_t , wage moves more $\Rightarrow$ stiff labor

$\Rightarrow$ small hour volatility. (each person chooses hours continuously)

Consistency with facts

$$n_t = \frac{H_t}{\bar{L}} = \frac{H_t}{L_t} \cdot \frac{L_t}{\bar{L}} \approx n^*(1 - u_t) = n^* \pi_t,$$

where u_t is the unemployment rate.

$$\frac{H_t}{L_t} = \bar{n}_t \approx n^* \quad \text{is empirically stable.}$$

*Cyclical movement in total hours comes from (extensive margin) employment, not from hours per worker (intensive margin).

Other RBC models

1. CRRA utility (King and Rebelo, 1999):

$$U(C_t, L_t) = \ln C_t + A \frac{(1 - N_t)^{1-\chi} - 1}{1 - \chi}.$$

Investment volatility $\uparrow\uparrow$ (almost correct), but underpredicts persistence, over-predicts comovement; consumption & labor are less volatile in model than data.

change preference curvature and more volatile TFP

2. KPR utility (King, Plosser and Rebelo, 1988):

$$U(C_t, L_t) = \frac{[C_t v(L_t)]^{1-\sigma} - 1}{1 - \sigma}.$$

Non-separable preference between C and N . $U_{C,t}$ depends on L_t , U_{L_t} depends on C_t . Relax the too-tight linkages $\Rightarrow$ improve comovement patterns.

3. GHH utility (Greenwood–Hercowitz–Huffman, 1988, AER)

$$U(C_t, N_t) = \frac{1}{1 - \sigma} \left[\left(C_t - A \frac{N_t^{1+\psi}}{1 + \psi} \right)^{1-\sigma} - 1 \right].$$

Cancel wealth effect, labor supply responds to substitution effect (wage/productivity), $\Rightarrow$ even more volatile N_t .

Since

$$-U_{N,t} = U_{C,t} W_t \quad \Rightarrow \quad A N_t^\psi = W_t.$$

4. Capital utilization:

$$Y_t = Z_t (u_t K_{t-1})^\alpha N_t^{1-\alpha},$$

$$K_t = I_t + [1 - \delta(u_t)] K_{t-1}, \quad \delta(u) > 0$$

Extra intensive margin for capital $\Rightarrow$ higher investment volatility, higher output volatility.